Name: _____ Date: _____

Year 9 Algebra: End-of-Topic Assessment

Section A: Algebraic Expressions and Index Laws

1. Simplify by collecting like terms: $7c + 2d - 3c + 6d$

2. Simplify by collecting like terms: $6y^2 + 2y - 2y^2 - 7y$

3. Simplify using multiplication: $5r \times 2s$

4. Simplify using multiplication and index laws: $4y^2 \times 3y^6$

5. Simplify using division: $\dfrac{24n^8}{6n^3}$

6. Expand: $5(2t - 3)$

7. Expand and simplify: $3(x+2)+4(2x-5)$

8. Factorise completely: $15x+20$

9. Factorise completely: $7p^2-21pq$

Section B: Substitution, Equations, and Inequalities

10. Given $a=5$, $b=-2$, and $c=4$, evaluate:

a) $3a-b$

b) c^2+ab

11. Solve the linear equation: $6x-5=31$

12. Solve the linear equation: $4(3k+2)=44$

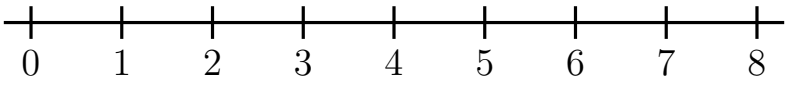
13. A plumber has a \$15 call-out fee and charges \$35 per hour on top of this.

a) Write an equation for the total cost, C, to hire him for h hours.

b) Use your equation to find how many hours the plumber worked if the total cost was \$190.

14. Solve the linear inequality: $5x + 2 \leq 22$

15. Represent the solution from Question 14 on the number line below. Draw your circle and arrow clearly.



Section C: Linear Patterns and Graphs

16. A pattern of matchsticks grows as shown in the table below:

Pattern number (n)	1	2	3	4
Number of matches (M)	5	9	13	17

a) Write a rule in words describing how the pattern grows.

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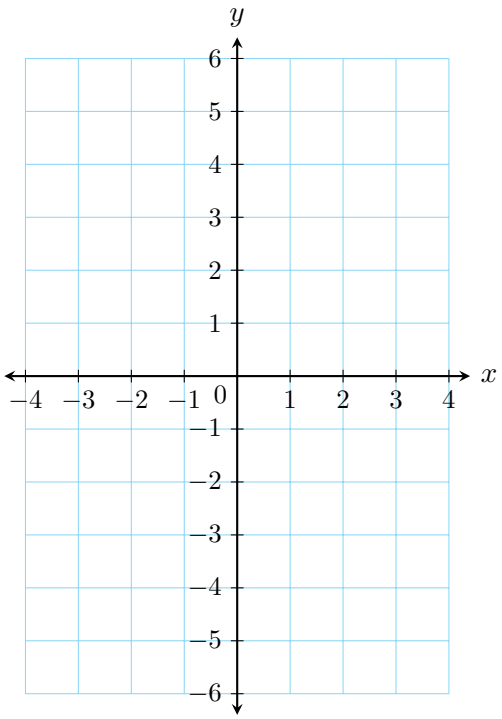
b) Write an equation linking M and n.

c) Use your equation to predict the number of matches needed for Pattern 30.

17. Complete the table of values for the linear relationship $y = 3x + 1$:

x	-2	-1	0	1
y				

18. Plot the points from Question 17 on the coordinate grid below and draw a straight line through them across all four quadrants.

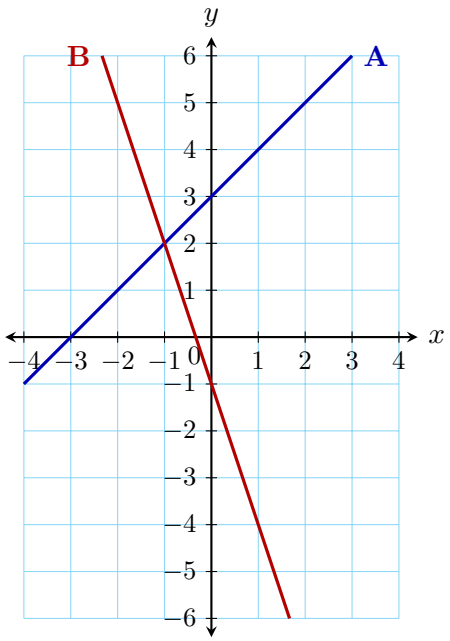


19. For the line $y = -2x + 6$:

a) State the gradient (m) and state whether it is positive, negative, or zero.

b) State the coordinates of the y-intercept.

20. Two straight lines, Line A and Line B, are drawn on the grid below.



a) Complete the table for each line.

Line	**Gradient (m)**	**Direction (uphill or downhill from left to right)**	**y-intercept**
A			
B			

b) Which line is steeper? Use the gradients to explain your answer.

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21. Jake says that $3x(2x + 1) = 6x^2 + 3x$. Lisa says that Jake is wrong and that expanding and equal are not the same.

Show, by substituting $x = 2$ and $x = -1$ into both expressions, that Jake is actually correct and both expressions are the same.

When $x = 2$:

When $x = -1$:

Marking grid

Cut scores

3b 0–5 3P: 6–10 3A: 11–15 4B: 16–22 4P: 23–29 4A: 30–35 5b: 36–41 5P: 42–47 5A: 47–50

Award 2 marks for straightforward questions and 3 marks for questions involving several steps or reasoning. The maximum marks for each question are shown below.

Question	Max marks	Marks awarded	Question	Max marks	Marks awarded
1	2		12	3	
2	2		13	3	
3	2		14	2	
4	2		15	2	
5	2		16	3	
6	2		17	2	
7	3		18	3	
8	2		19	2	
9	2		20	3	
10	3		21	3	
11	2				
Total	**50**				

Answers

1. $4c + 8d$

2. $4y^2 - 5y$

3. $10rs$

4. $12y^8$

5. $4n^5$

6. $10t - 15$

7. $3x + 6 + 8x - 20 = 11x - 14$

8. $5(3x + 4)$

9. $7p(p - 3q)$

10. a) $3(5) - (-2) = 15 + 2 = 17$
b) $4^2 + (5)(-2) = 16 - 10 = 6$

11. $6x = 36 \implies x = 6$

12. $12k + 8 = 44 \implies 12k = 36 \implies k = 3$ (or $3k + 2 = 11 \implies k = 3$)

13. a) $C = 35h + 15$
b) $35h + 15 = 190 \implies 35h = 175 \implies h = 5$ hours

14. $5x \leq 20 \implies x \leq 4$

15. A solid (closed) circle at 4 with an arrow pointing to the left along the number line.

16. a) Start with 5 matches and add 4 matches for each new pattern (or: multiply the pattern number by 4 and add 1).
b) $M = 4n + 1$
c) $M = 4(30) + 1 = 120 + 1 = 121$ matches

17. When $x = -2$, $y = -5$; when $x = -1$, $y = -2$; when $x = 0$, $y = 1$; when $x = 1$, $y = 4$.

18. Points $(-2, -5)$, $(-1, -2)$, $(0, 1)$ and $(1, 4)$ plotted accurately, with a single straight line drawn through them and extended across the grid.

19. a) Gradient $m = -2$; it is negative.
b) $(0, 6)$

20. a) Line A: gradient $m = 1$, uphill (positive), y-intercept $(0, 3)$.
Line B: gradient $m = -3$, downhill (negative), y-intercept $(0, -1)$.
b) Line B is steeper, because the size of its gradient (3) is greater than the size of Line A's gradient (1). The negative sign only tells us the direction, not the steepness.

21. When $x = 2$: $3(2)(2 \times 2 + 1) = 6 \times 5 = 30$ and $6(2)^2 + 3(2) = 24 + 6 = 30$.
When $x = -1$: $3(-1)(2 \times (-1) + 1) = (-3)(-1) = 3$ and $6(-1)^2 + 3(-1) = 6 - 3 = 3$.
Both expressions give the same value each time, so Jake is correct: expanding gives an equivalent (equal) expression.